

**Lemma.** Let  $\mathbb{Q}: F \rightarrow F'$  be a field Isomorphism, and let  $\mathbb{Q}^*: F[x] \rightarrow F'[x]: \sum r_i x^i \mapsto \sum \mathbb{Q}(r_i) x^i$ .  
 Let  $p(x) \in F[x]$  be irreducible, and let  $p^*(x) = \mathbb{Q}^*(p(x))$ .  
 If  $\beta$  is a root of  $p(x)$  and  $\beta'$  is a root of  $p^*(x)$ ,  
 then, there exists a unique Isomorphism  $\overline{\mathbb{Q}}: F(\beta) \rightarrow F(\beta')$  s.t.  $\overline{\mathbb{Q}}|_F = \mathbb{Q}$ ,  $\overline{\mathbb{Q}}(\beta) = \beta'$ .  
 Proof. Since  $\mathbb{Q}^*$  is also Isomorphism,

$$\mathbb{I}: F[x]/(p(x)) \rightarrow F[x]/(p^*(x)): f(x) + (p(x)) \mapsto \mathbb{Q}^*(f(x)) + (p^*(x))$$

It maps  $c + (p(x)) \mapsto \mathbb{Q}(c) + (p^*(x))$  for all  $c \in F$  and  $x + (p(x)) \mapsto x + (p^*(x))$ , and Isomorphism.

Now,  $\overline{\mathbb{Q}}: F(\beta) \xrightarrow{\phi^{-1}} F[x]/(p(x)) \xrightarrow{\mathbb{I}} F[x]/(p^*(x)) \xrightarrow{\phi'} F(\beta')$

$$: \alpha \mapsto (\phi' \circ \mathbb{I} \circ \phi^{-1})(\alpha)$$

'is a field isomorphism leaving  $F$  fixed, and  $\beta$  maps to  $\beta'$ .

**Lemma.** Let  $f(x) \in F[x]$  and  $D_x f(x) \in F[x]$  be a derivative of  $f(x)$ . Then,

$f(x)$  has a multiple root  $\alpha$  'if and only if'  $D_x f(x)$  has  $\alpha$  as a root.

In particular,

$f(x)$  is separable 'if and only if'  $(f(x), D_x f(x)) = 1$ .

Proof. Suppose that  $(x-\alpha)^2$  divides  $f(x)$  in the splitting field  $K$  of  $f$ . Then,

$$f(x) = (x-\alpha)^2 \cdot h(x) \quad \text{for some } h(x) \in K[x].$$

By the property of derivative,

$$D_x f(x) = 2 \cdot (x-\alpha) \cdot h(x) + (x-\alpha)^2 \cdot D_x h(x) = (x-\alpha)(2h(x) + (x-\alpha) \cdot D_x h(x)).$$

Thus proved. Conversely, suppose that both  $f$  and  $D_x f$  has  $\alpha$  as a root. Then,

$$f(x) = (x-\alpha) \cdot h(x) \quad \text{for some } h(x) \in K[x], \text{ and } D_x f(x) = h(x) + (x-\alpha) \cdot D_x h(x).$$

And, by the assumption,  $(x-\alpha)$  divides  $D_x f$ , so divides  $h(x)$ . Now,

$$f(x) = (x-\alpha)h(x) = (x-\alpha)^2 r(x), \quad \text{where } r(x) = h(x)/(x-\alpha).$$

Now, the last assertion 'is obtained by',

$f$  is separable 'if and only if'  $f$  has no multiple root 'if and only if'  $\deg(f(x), D_x f(x)) = 0$  'if and only if'  $(f(x), D_x f(x)) = 1$ .

Thm. A field of characteristic 0 is perfect.

Proof. Let  $f(x) \in F[x]$  be an irreducible polynomial. If  $D_x f(x)$  is not zero, then  $\deg D_x f < \deg f$ .

Therefore,  $(f, D_x f) = 1$ , because if  $(f, D_x f) \neq 1$ , then it is contradiction to  $f$  is irreducible. So it is separable.

Lemma. A field of characteristic  $p$  is perfect iff every element of  $F$  has a  $p$ th root in  $F$ .

Proof. Let  $F$  be a field of characteristic  $p$ .

Suppose that  $F$  is perfect. Let  $\alpha \in F$  be a non-zero, non-1. put  $f(x) = x^p - \alpha$ .

If  $f(x)$  has a root in  $F$ , then there is nothing to prove.

Assume  $f$  has no root in  $F$ . Observe that in splitting field  $K$  of  $f(x)$ , put  $\beta \in K$  s.t.

$$f(\beta) = \beta^p - \alpha = 0.$$

And,  $f(x) = x^p - \alpha = x^p - \beta^p = (x - \beta)^p$  in  $K$ . But, if  $f$  is reducible in  $F$ , then for some  $1 \leq k < p$ ,  $f(x) = (x - \beta)^{p-k} \cdot (x - \beta)^k$  and  $(x - \beta)^{p-k}, (x - \beta)^k \in F[x]$ .

That is,  $(-\beta)^k \in F$ . Clearly,  $(k, p) = 1$ , thus  $ku + pv = 1$  for some  $u, v \in \mathbb{Z}$ .

Now,  $\beta = \beta^{ku+pv} = (\beta^k)^u \cdot (\beta^p)^v = (\beta^k)^u \cdot \alpha^v \in F$ . It is contradiction. But, if  $f$  is irreducible then  $D_x f = p \cdot x^{p-1} = 0$  implies contradiction with  $F$  is perfect. Now,  $\alpha$  has a  $p$ th root in  $F$ .

Conversely, let  $f(x) \in F[x]$  be an irreducible. Assume it is inseparable.

Denote  $f(x) = a_n x^n + \dots + a_1 x + a_0$ . Since  $f(x)$  is irreducible and inseparable,

$D_x f(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \dots + a_1$  must be zero, that is,

$$a_k \neq 0 \Leftrightarrow k \in (p)$$

Therefore, we can write

$$f(x) = b_m x^{mp} + b_{m-1} x^{(m-1)p} + \dots + b_1 x^p + b_0$$

Now, by the assumption, each  $b_i$  has  $p$ th root  $c_i$ , thus

$$f(x) = c_m^p \cdot (x^m)^p + c_{m-1}^p \cdot (x^{m-1})^p + \dots + c_0^p$$

$$= (c_m \cdot x^m + c_{m-1} \cdot x^{m-1} + \dots + c_0)^p. \text{ But it is reducible. Contradiction.}$$

Corollary. Every finite field is perfect. (char is  $p$  on finite thus Auto.)

Thm. Every finite separable extension is simple extension.

Proof. If  $F$  is finite, then the finite extension  $E/F$  is also finite. Then,  $E^X$  is cyclic, therefore  $E = F(\alpha)$  where  $\alpha$  is generator of  $E^X$ .

Now, assume that  $F$  is infinite. Since  $E/F$  is finite,

$$E = F(\alpha_1, \dots, \alpha_r) \text{ where } \alpha_1, \dots, \alpha_r \in E.$$

Now, enough to show  $F(\beta, \gamma) = F(\alpha)$  for some  $\alpha \in E$ , and after is solved by the induction.

Let  $\beta_1, \dots, \beta_n$  be distinct zeros of  $\text{irr}(\beta, F)$ , and

$\gamma_1, \dots, \gamma_m$  be distinct zeros of  $\text{irr}(\gamma, F)$ , in  $\bar{F}$ . ( $\beta_1 = \beta$ ,  $\gamma_1 = \gamma$ ).

Since  $E/F$  is separable, each zeros are multiplicity of 1.

Take  $a \in F$  s.t.  $a \neq \frac{\beta_i - \beta}{\gamma - \gamma_j}$  for all  $1 \leq i \leq n$ ,  $2 \leq j \leq m$ , since  $F$  is infinite.

Put  $\alpha = \beta + a \cdot \gamma$ . Then, for all  $(i, j) \in \{1, \dots, n\} \times \{2, \dots, m\}$ ,

$$\alpha = \beta + a \gamma \neq \beta_i + a \gamma_j, \text{ or } \alpha - a \gamma_j \neq \beta_i.$$

Observe that:  $\text{irr}(\beta, F)(\alpha - a\gamma) \in F(\alpha)[x]$ , and

$$\text{irr}(\beta, F)(\alpha - a\gamma) = \text{irr}(\beta, F)(\beta) = 0.$$

But,  $\text{irr}(\beta, F)(\alpha - a\gamma_j) \neq 0$  for all  $2 \leq j \leq m$ ,

That is,  $\text{irr}(\beta, F)(\alpha - a\gamma)$  and  $\text{irr}(\beta, F)(x)$  have a common factor  $(x - \gamma)$  in  $F(\alpha)$ , thus  $\gamma \in F(\alpha)$ , so  $\beta = \alpha - a\gamma \in F(\alpha)$ . ~~so~~